

DIAL-ALERT

Step 9: Use of Generative AI

Optional capstone addendum | Reviewed 18 September 2026

1. How Generative AI was used

A language-model authoring assistant drafted a plain-language summary of aggregate cohort and model results, and assisted with validation code, tests, documentation and the demonstration script. The tool disclosure is recorded in the accompanying README. The exact model identifier and generation settings were not recorded.

This is an authoring-assistance use case. The executable code replays a saved AI draft with checked numbers; it does not perform a fresh LLM call. The random forest remains the predictive model.

2. Workflow and source evidence

Frozen model card and test metrics -> curated aggregate facts -> AI-drafted text with named placeholders -> numeric validation -> rendered summary -> investigator review.

Sources are evidence/model_card.md, evidence/original_test_metrics.json and evidence/decision_threshold.json. No patient-level data are needed. The source artifacts are inherited from earlier project work; raw-data analysis and training were not rerun for Step 9.

3. Example output

The analytic cohort contains 106,758 dialysis sessions from 830 patients. The recorded event prevalence is 8.49%. Events are therefore less common than non-events, so accuracy alone would be an incomplete description of prediction performance.

The held-out test set contains 21,354 sessions from 170 patients. Existing test results include average precision 0.395, ROC AUC 0.852 and Brier score 0.062. These are retrospective model results, not evidence of improved patient outcomes.

4. Reproduce the saved example

From the repository root:

```
python3 step9_genai/src/replay_summary.py
```

```
python3 -m unittest discover -s step9_genai/tests -v
```

Python 3.10 or newer; standard library only for replay and tests. Video regeneration additionally requires Pillow and ffmpeg. No API key or network connection is required.

Verification and limitations

5. Included materials

Documentation: README.md and this PDF. Code: src/replay_summary.py. Examples: examples/aggregate_facts.json, examples/ai_draft.json and examples/rendered_summary.md. Demo: demo/DIAL_ALERT_Step9_Demo.mp4.

The 63-second silent, captioned video presents actual recorded local output across seven scenes. It is a rendered walkthrough, not a browser screencast or recording of live LLM generation.

6. Verification and limitations

The original eight summary tests verify numerical consistency and reject changed metrics, cohort counts, unknown fields, invented numeric literals and unsupported placeholders. Additional publication tests guard the reviewed file inventory, hashes and selected secret/contact and dialogue patterns.

Numeric validation cannot judge arbitrary clinical claims. The tests explicitly demonstrate that unsupported nonnumeric prose may pass. Investigator review remains pending. Fresh language generation cannot be reproduced from the saved example; there is no live chatbot, recommender or external LLM integration.

7. Privacy review

The review covered every original Step 9 file, all seven video scenes and media metadata. No private dialogue, authoring instructions, credentials, contact details or patient records were found. The personal author byline was removed from the current README. This PDF contains no personal author metadata.

The intentional AI-use disclosure remains necessary for the academic task. Only aggregate study facts and final project artifacts are published. Automated scans complement review; they do not guarantee detection of every possible secret. Earlier Git history and public account/commit attribution remain visible.